MineAI ? All-in-One AI Productivity Hub

Complete System Architecture, Feature Workflows, Database Schemas & RAG Knowledge Base

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1. Executive Summary & Application Overview

MineAI is an enterprise-grade personal AI productivity super-app designed to unify daily digital workflows, multi-store smart shopping, localized information ingestion, secure credentials storage, and personal wellness into a single mobile experience.

- **Target Platforms: Native Android and iOS cross-platform mobile app built on Flutter** 3.29+, complemented by an administrative Flutter web dashboard and a Node.js Express 5 microservice backend.
- **Core Philosophy: Privacy-first architecture with localized offline SQLite storage,** biometric security, client-side encryption, and cloud synchronization powered by Supabase Auth and PostgreSQL.
- **AI Integration: Context-aware task extraction, automated professional email authoring,** weather advisory insights, and integrated Flowise AI RAG chat assistance.
- **Accessibility & Localization: Clamped TextScaler.linear(1.0) preventing UI overflow** across all system font scaling configurations, alongside full dual-language localization (English and Tamil).

2. System Architecture & Tech Stack

The MineAI platform consists of three core codebases working collaboratively:

- **Flutter Frontend (Client App): Located in flutter frontend/. Built with BLoC/Cubit** state management, GoRouter declarative routing, Dio HTTP client with Alice traffic inspector, and dynamic theme switching across 5 distinct visual palettes.
- **Node.js Backend (REST API): Located in Backend/. Express 5 microservice engine** communicating with Supabase PostgreSQL, Google Generative AI (Gemini 2.5 Flash), Cheerio web scraping, Nodemailer SMTP transport, and Playwright headless automation.
- **Admin Portal: Located in admin/. Flutter Web/Desktop control center providing real-time** system metrics, user tier audits, support ticket resolution, and OTT catalog curation.

Technical Specifications Table

Frontend Framework:	Flutter 3.29+ / Dart 3.8+ (Clean Layered Architecture)
State Management:	flutter_bloc 9.1.1 & Cubit Patterns
Networking Client:	Dio 5.8.0 with interceptors & Alice inspector
Cloud Backend:	Node.js 18+, Express 5.2.1, ESM Architecture
Database & Auth:	Supabase PostgreSQL (Cloud) + SQLite (Local Device)
AI Model Engines:	Google Gemini 2.5 Flash via @google/genai & Flowise RAG
Encrypted Storage:	flutter_secure_storage + AES-256 for Vault credentials
Media & Streaming:	HLS adaptive video player (Auto, 720p, 480p, 360p)

3. Detailed Core Features & Workflows

Feature 1: AI Voice & Speech Task Extraction

Converts spoken audio or conversational voice notes into structured, scheduled tasks with automated field parsing.

- **Workflow: User records audio via speech_to_text. Audio transcript is submitted to POST** /api/v1/tasks/extract. The Gemini LLM parses task title, description, category (Work, Personal, Urgent, Meeting), priority (Low, Medium, High), and due dates/times.
- **Storage & Alarms: Parsed tasks are saved into local SQLite database. Local** notifications are scheduled via flutter_local_notifications at the configured alarm timestamp.
- **Offline Resilience: If offline or if AI token quota is exceeded, voice audio** transcription can be saved directly as a raw task draft without LLM parsing.

Feature 2: Curated City News Web Scraper

Delivers verified, localized daily news feeds extracted directly from national publishers with zero algorithmic sensationalism.

- **Data Sources: Live RSS feeds from The Hindu and Times of India, coupled with Cheerio** web scraping for high-definition editorial photography.
- **Boundary Filtering: Strict city boundaries enforcing content relevance for Chennai,** Bengaluru, Mumbai, Delhi, Hyderabad, Kolkata, Coimbatore, and Madurai.
- **Features: Publisher filter chips, article summaries, direct in-app browser article** viewer, and native system share sheet integration.

Feature 3: Real-Time Multi-Store Price Comparison

Automates product price intelligence across major Indian e-commerce marketplaces to guarantee users get the lowest price.

- **Marketplaces Scraped: Amazon India, Flipkart, and Meesho.**
- **Engine: Backend runs parallel scraping via multiStoreScraper.service.js. Normalizes** prices, stock status, star ratings, and store logos.
- **Deal Detection: Calculates percentage savings between the highest and lowest price.** Highlights 'Best Value' badges and provides direct store deep links.
- **Saved Deals: Allows bookmarking products into Supabase saved_deals table for ongoing** price drop tracking.

Feature 4: AI Email Assistant & Direct SMTP Dispatch

An automated communications suite for drafting, polishing, and dispatching high-impact emails.

- **Generation Modes: Accepts topic or bullet points and generates emails across 4 tones:** Professional, Casual, Friendly, and Urgent.
- **Direct Dispatch: Users can link custom Gmail App Passwords or standard SMTP credentials** to send generated emails immediately from the app without leaving MineAI.
- **Cloud Templates: Save generated emails under custom labels and folders in Supabase** email_templates.

Feature 5: Encrypted Private Vault

A hardened, zero-knowledge biometric safe for storing sensitive credentials and confidential records.

- **Security Model: Protected by biometric fingerprint/FaceID authentication via** local_auth. Credential secrets are encrypted client-side using AES-256 and stored in flutter_secure_storage.
- **Item Types: Website logins, credit/debit card numbers, Wi-Fi keys, secure notes, bank** account identifiers, and confidential documents.
- **Privacy: Vault items never transmit plaintext credentials to cloud servers; secrets** remain exclusively on the user device.

Feature 6: Personal Diary & Mood Celebrator

A daily reflective journaling system combining multimedia memory logs with mood wellness tracking.

- **Features: Supports rich text notes, audio voice memo recordings, and gallery photos.** Tracks daily emotional mood scores (1 to 5 stars).
- **Mood Poster: Renders an artistic 'Mood Vibe' poster snapshot ready for instant export** and social media sharing.
- **Storage: Entries are stored in SQLite diary_entries with optional cloud sync to** Supabase.

Feature 7: Expense & Budget Tracker

Comprehensive personal finance manager with visual spend distribution and financial statements.

- **Features: Logs income and expenses with customizable categories (Food, Bills, Travel,** Shopping, Health). Visual budget progress bar warns of limit overruns.
- **Reporting & Export: Generates itemized financial reports with direct one-tap export to** Excel (.xlsx) and printable PDF statements via the native pdf library.

Feature 8: MineAI Streams (OTT Video Hub)

A curated media streaming player supporting modern HLS (HTTP Live Streaming) adaptive bitrate playback.

- **Playback Controls: Multi-quality switcher (Auto, 720p, 480p, 360p), playback speed from** 0.5x to 2.0x, aspect ratio toggles, and swipe gesture controls for volume and brightness.

Feature 9: Weather Insights & AI Recommendations

Real-time meteorological forecasts coupled with intelligent daily lifestyle recommendations.

- **Data Provided: Temperature, humidity, UV index, wind speed, and precipitation chance.**
- **AI Advice: Gemini generates personalized contextual advice (e.g. workout suitability,** umbrella reminders, optimal commute times).

Feature 10: BYOK Custom API Key Management

Empowers advanced users to connect their personal Google Gemini or OpenAI API keys.

- **Benefit: Bypasses default free tier monthly token quotas with zero rate limits. Stored** securely with asymmetric encryption.

Feature 11: Support Ticket Helpdesk

Integrated customer care portal allowing users to raise support inquiries directly within the app.

- **Statuses: Open -> In Progress -> Resolved -> Closed. Synchronized in real time with the** Flutter Admin dashboard.

Feature 12: Dynamic Theming & Localization

User-customizable look-and-feel designed for all lighting conditions and multilingual users.

- **Color Modes: Deep Dark, Pure Light, Natural, Orange, and White-Blue.**
- **Languages: Full English (en) and Tamil (ta) localization powered by easy_localization.**

4. Database Architecture & Schemas

MineAI uses a hybrid storage model: Supabase PostgreSQL for cloud sync and user management, alongside local SQLite databases for instant offline access.

Supabase PostgreSQL Tables

- **profile: id (UUID PK), full_name, email, avatar_url, plan_id (free/pro/premium),** token_balance, theme_mode, language, created_at, updated_at.
- **user_keys: id (UUID PK), user_id (FK), provider (gemini/openai), encrypted_key,** is_active, created_at.
- **tickets: id (UUID PK), user_id (FK), subject, description, priority** (low/medium/high/urgent), status (open/in_progress/resolved/closed), admin_notes, created_at.
- **email_templates: id (UUID PK), user_id (FK), title, subject, body, tone, label,** created_at.
- **saved_deals: id (UUID PK), user_id (FK), product_title, store_name, price,** original_price, product_url, image_url, created_at.

Local SQLite Tables (Client Device)

- **tasks: id TEXT PRIMARY KEY, user_id TEXT, title TEXT, description TEXT, category TEXT,** priority TEXT, due_date TEXT, due_time TEXT, is_completed INTEGER, alarm_enabled INTEGER.
- **expenses: id TEXT PRIMARY KEY, user_id TEXT, title TEXT, amount REAL, type TEXT** (income/expense), category TEXT, date TEXT, payment_method TEXT, notes TEXT.
- **budgets: id TEXT PRIMARY KEY, user_id TEXT, category TEXT, monthly_limit REAL,** month_year TEXT.
- **diary_entries: id TEXT PRIMARY KEY, user_id TEXT, title TEXT, content TEXT, mood_score** INTEGER, photo_path TEXT, audio_path TEXT, created_at TEXT.

5. Backend REST API Reference

All endpoints are prefixed with /api/v1 and require an Authorization: Bearer <Supabase_JWT>

header unless designated public.

Core Endpoints Listing

- **POST /api/v1/auth/signup: Create new account with email & password.**
- **POST /api/v1/auth/login: Authenticate user and return JWT session token.**
- **GET /api/v1/user/me: Retrieve user profile, active plan tier, and token quota.**
- **PUT /api/v1/user/profile: Update full name, avatar, and app preferences.**
- **POST /api/v1/tasks/extract: Accepts audio transcription text; returns parsed task JSON.**
- **GET /api/v1/prices/search?q=<query>: Multi-store scraper querying Amazon, Flipkart,** Meesho.
- **POST /api/v1/email-helper/generate: Generates email copy with requested tone and** length.
- **POST /api/v1/email-helper/send: Dispatches email via user-configured SMTP/Gmail.**
- **GET /api/v1/news/feed?city=<city>: Fetches city-filtered news articles with editorial** photos.
- **GET /api/v1/weather/current?city=<city>: Fetches live weather and Gemini health advice.**
- **POST /api/v1/tickets: Submits support ticket with priority and description.**
- **POST /api/v1/users/api-keys: Stores encrypted custom Gemini/OpenAI API key.**

6. Flowise AI RAG Implementation Guide

To build the RAG (Retrieval-Augmented Generation) chatbot for MineAI, follow this exact architecture inside Flowise AI:

Step 1: Install & Start Flowise

Install Flowise globally or run it via npx / Docker:

```
# Option A: Using NPX directly
npx flowise start

# Option B: Using Global NPM Install
npm install -g flowise
flowise start --PORT 3000

# Flowise UI will open at: http://localhost:3000
```

Step 2: Flowise Chatflow Canvas Architecture

Create a new Chatflow named 'MineAI Assistant' and drag these nodes onto the canvas:

- **1. Document Loader Node: Select 'PDF File Loader'. Upload this generated MineAI** Documentation PDF.
- **2. Text Splitter Node: Select 'Recursive Character Text Splitter'. Configure Chunk Size** = 1000, Chunk Overlap = 200, Separators = ['\n\n', '\n', ' ', ''']. Connect output to Document Loader's Text Splitter input.
- **3. Embeddings Node: Select 'Google GenerativeAI Embeddings' (Model: text-embedding-004)** or 'OpenAI Embeddings' (Model: text-embedding-3-small). Enter your API Key.
- **4. Vector Store Node: Select 'In-Memory Vector Store' (for local testing) or 'Supabase** Vector Store' (production). Connect Document Loader to Vector Store's Document input, and Embeddings to Vector Store's Embeddings input.
- **5. Chat Model Node: Select 'ChatGoogleGenerativeAI' (Model: gemini-1.5-flash) or** 'ChatOpenAI' (Model: gpt-4o-mini). Set Temperature = 0.2 for precise, factual answers.
- **6. Memory Node: Select 'Buffer Memory' or 'Conversation Summary Memory'. Set Session ID** key = sessionId.
- **7. Retrieval QA Chain Node: Select 'Conversational Retrieval QA Chain'. Connect Chat** Model, Vector Store (as Retriever), and Memory. Save and Deploy!

Step 3: System Prompt Configuration

Insert this prompt into the Chatflow System Prompt parameter:

```
You are the official MineAI AI Assistant. You assist users with all questions
about the MineAI mobile application, its features, workflows, and settings.
Always base your responses on the retrieved MineAI documentation context.
Be friendly, polite, concise, and format answers using Markdown bullet points.
If a user asks about an unsupported feature, guide them to contact Support Tickets.
```

Step 4: Flowise REST API Integration

Flowise exposes a prediction endpoint that the Flutter app queries:

```
POST http://localhost:3000/api/v1/prediction/<YOUR_CHATFLOW_ID>
Content-Type: application/json

{
  "question": "How do I compare prices between Amazon and Flipkart?",
  "chatId": "unique-user-session-id",
  "streaming": false
}
```

7. Comprehensive FAQs & Knowledge Base Q&A

These verified Q&A pairs are indexed into vector embeddings for instant RAG lookup:

- **Q: How does Voice Task Extraction work in MineAI? A: Go to Voice Tasks on the Home** screen, tap the microphone button, and speak your thoughts naturally (e.g., 'Schedule meeting with Rahul tomorrow at 3 PM'). MineAI uses speech recognition and Gemini AI to automatically parse the title, date, time, priority, and category, then saves it to your local task list with notification reminders.
- **Q: Which stores are supported in Real-Time Price Comparison? A: MineAI searches and** compares prices in real time across Amazon India, Flipkart, and Meesho. It displays ratings, stock status, and calculates exact price differences to highlight the best deal.
- **Q: Is my data in the Private Vault encrypted? A: Yes. All passwords, bank details, and** notes in the Secure Vault are encrypted client-side using industry-standard AES-256 encryption and stored in hardware-backed secure storage. Biometric authentication (fingerprint/FaceID) is mandatory to unlock.
- **Q: How do I send emails using the AI Email Helper? A: Navigate to AI Email Helper,** enter a quick prompt or bullet points, and choose a tone (Professional, Casual, Friendly, Urgent). Once generated, you can edit it and send it directly via your linked Gmail App Password or SMTP account, or save it as a template.
- **Q: Which cities are supported in Local City News? A: MineAI currently provides** localized RSS feeds and editorial photos for 8 major Indian metropolitan cities: Chennai, Bengaluru, Mumbai, Delhi, Hyderabad, Kolkata, Coimbatore, and Madurai.
- **Q: Can I export my expense tracking reports? A: Yes. In the Expense Tracker screen, tap** the Export button to export itemized monthly transaction reports directly into Excel (.xlsx) spreadsheets or printable PDF statements.
- **Q: What should I do if I run out of free AI tokens? A: You can either upgrade to a Pro** plan in the Subscription screen, or go to Settings -> API Keys and enter your personal Google Gemini or OpenAI API key (BYOK) for unlimited zero-cost generation.
- **Q: How do I change the app theme and language? A: Open Settings from the Profile tab.** Under Appearance, select from 5 themes (Deep Dark, Pure Light, Natural, Orange, White-Blue). Under Language, toggle between English and Tamil.